

12(a)	<i>either</i>	
	(minimum) energy required / work done to separate the nucleons (in a nucleus)	M1
	to infinity	A1
	<i>or</i>	
	energy released when nucleons come together (to form a nucleus)	(M1)
	from infinity	(A1)
12(b)(i)	(total) binding energy of thorium and helium (nuclei) greater than binding energy of uranium (nucleus)	B1
12(b)(ii)1	change in mass = $238.05076 - (234.04357 + 4.00260)$ = $4.59 \times 10^{-3} \text{ u}$	A1
12(b)(ii)2	<i>either</i>	
	$E = mc^2$ = $4.59 \times 10^{-3} \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$	C1
	= $6.9 \times 10^{-13} \text{ J}$	A1
	<i>or</i>	
	1u = 931 MeV $E = 4.59 \times 10^{-3} \times 931 \times 10^6 \times 1.6 \times 10^{-19}$	(C1)
	= $6.8 \times 10^{-13} \text{ J}$	(A1)
12(b)(iii)	Th nucleus / He nucleus / product nucleus has kinetic energy	M1
	energy of gamma photon must be less than energy released	A1